

Home Plant Health Prediction Application With Cellphone Camera And AI Based On Federated Learning

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ABSTRACT

Accurately monitoring the health of house plants is becoming increasingly important as public interest in urban farming and ornamental plants increases. This research develops a plant health prediction application using cell phone cameras and artificial intelligence (AI) based on federated learning. This technology enables AI model training directly on the user's device without having to send image data to a central server, thus maintaining user privacy and network efficiency. The app uses plant image classification models to detect disease symptoms such as leaf spot, wilting, and discoloration. Data is collected from various users, and the model is trained in a decentralized manner through a federated averaging protocol. Test results show that the system is able to achieve prediction accuracy above 85% in detecting plant conditions, with low latency and minimal data consumption. This approach shows significant potential in the application of secure, distributed, and user-friendly AI for home plant monitoring.

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1. INTRODUCTION

In recent years, public interest in ornamental plants and household plants has increased significantly (Andri & Tumbuan, 2015). This phenomenon not only occurs in urban areas as a form of spatial aesthetics, but also as part of a healthy and environmentally friendly lifestyle. House plants such as monstera, betel nut, cactus, and various varieties of succulents are now an important part of interior decoration as well as homeowners' activities in maintaining air quality and reducing stress. However, not all plant owners have sufficient knowledge on how to optimally care for their plants, especially in recognizing the early signs of plant stress, pest attack, or disease (Kampus & Dramaga, n.d.).

With advances in mobile device technology, almost everyone now has access to high-resolution cameras and internet connectivity. This opens up opportunities to develop visual and artificial intelligence-based applications that can be used directly on the user's device, without the need for additional tools or high costs. The utilization of mobile phone cameras as visual monitoring tools allows the process of identifying disease symptoms in plants to be done quickly, practically, and without special skills. However, processing large amounts of plant image data raises concerns about privacy, data security, and computational efficiency. The Federated Learning approach is a strategic solution because it enables collaborative AI model training among user devices without the need to send raw data to a central server. With FL, AI models can be continuously improved in accuracy by utilizing the collective experience of many users, without compromising the privacy of their personal data.

Leaf discoloration, spotting, or unnatural growth are often ignored until they cause permanent damage. Therefore, a support system is needed that is able to assist home users in monitoring plant health conditions quickly, easily, and accurately (Latifah et al., 2021). Plant health is an important factor in the success of cultivation, both in the large-scale agricultural sector as well as in household-scale

ornamental and consumer crops. In the midst of increasing public interest in urban farming and independent farming in the home environment, the challenge of taking optimal care of plants is becoming increasingly apparent. One of the main challenges is the limited knowledge and ability to detect symptoms of plant diseases early on, which often leads to sudden plant damage or death.

Along with the rapid development of digital technology and artificial intelligence (AI), there is an opportunity to utilize the smartphone camera that is now almost owned by everyone as the main tool in detecting plant conditions. Mobile phone cameras not only function as documentation tools, but also as visual sensors that can capture various visual symptoms in plants directly. The use of personal data, including photos of plants taken by users, raises privacy concerns. Many applications that rely on centralized data collection risk exposing user information without permission. To address this challenge, the Federated Learning (FL) approach comes as an innovative solution that allows AI model training to be done directly on the user's device without sending data to a central server. The model only sends parameter updates, not raw data, so user privacy is maintained.

The development of artificial intelligence (AI) technology offers innovative solutions in detecting and predicting plant health conditions. By utilizing the camera on smartphones, users can now take pictures of plants and get an automated analysis of their condition. However, behind this convenience, there are new challenges related to data privacy and limited computing infrastructure on user devices. Federated Learning (FL) emerges as a smart approach to overcome these challenges. Unlike conventional AI approaches that require data collection to a central server, FL allows AI model training to be done in a distributed manner across user devices. Thus, the user's personal data, including plant images and environmental metadata, remains secure and does not have to leave the device.

The health of houseplants is an increasing concern in modern urban life, where plants not only serve as decorative elements, but also provide psychological benefits, improve air quality, and create a healthier living environment. However, many house plant owners do not have sufficient knowledge or skills in early detection of disease or stress in plants. In this context, artificial intelligence (AI) technology based on computer vision through smart phone cameras can be a practical and affordable solution in detecting plant health automatically and in real-time. AI-based crop health prediction applications have actually developed rapidly in recent years, but most of the research focuses on large-scale agricultural crops and uses centralized datasets (Mahendra et al., 2024). This approach is not fully suited to the household context, which has more varied crop types, uncontrolled shooting environments, and user privacy limitations. In addition, most conventional machine learning models require data to be collected and sent to a central server, which potentially raises privacy and security concerns for user data (Sudirwo et al., 2025).

Federated Learning (FL) comes as a new approach in training AI models without having to send raw data to a central server. In this context, models are trained locally on the user's device (e.g., mobile phone), and only model parameters are shared with the central server for combination. This approach offers advantages in terms of privacy protection and communication efficiency. Therefore, the application of FL in HP camera-based house plant health prediction applications is a very relevant and significant innovation (Besar, n.d.).

This research is important because it fills a gap in the literature that is still minimal regarding the application of Federated Learning for plant health diagnosis in the household environment. Most previous studies have not specifically integrated FL technology with cell phone cameras in the context of house plants, which have very diverse visual and environmental characteristics. In addition, this research also responds to the need for a solution that is inclusive, personalized, and practical for non-expert users in detecting plant diseases. The main focus of this research is on the development of a home plant health prediction application that can be accessed through a mobile phone camera by utilizing AI technology based on Federated Learning (Nirsal et al., 2025). This application will be able to detect the early symptoms of plant diseases based on images taken with a cell phone camera.

This research has three main contributions: (1) developing a prototype AI-based mobile application capable of predicting the health conditions of house plants using a cell phone camera, (2) implementing a Federated Learning architecture to maintain user data privacy and improve system efficiency, and (3) evaluating system performance in real-world scenarios with variations in plant types and lighting conditions.

2. RESEARCH METHOD

This research belongs to the type of applied research that aims to present concrete solutions to real problems faced by home users in caring for plants, especially in terms of early detection of plant diseases. By utilizing highly accessible mobile phone camera technology combined with artificial intelligence (AI) and Federated Learning (FL) approaches, this research offers relevant and applicable innovations amidst the increasing public interest in home farming and urban farming. The main target population is home gardeners who use smartphone devices to care for and monitor the condition of their plants. This population includes individuals of various ages, professions, and levels of horticultural knowledge in both urban and suburban environments who have an interest in small-scale ornamental, medicinal, or food crops grown in yards, balconies, or indoors. This population was selected based on its main characteristic, the use of smartphones as the main device for plant documentation and monitoring. This is in line with the research objective of developing a home plant disease detection application based on cell phone cameras and AI, so users from this group are representative of the actual usage scenario. Data collection in this study was conducted through a combined approach of technology-based direct observation, digital documentation, and user questionnaires, with the main focus on capturing images of plants through a cell phone camera as well as user interaction with the developed application. These techniques were designed to generate two main types of data: visual image data of plants and behavioral/interaction data of users towards the system. The main technique for data collection was image capturing done directly by respondents using their mobile phone cameras. In this study, the data analysis techniques were quantitative and experimental, focusing on processing the plant image data, evaluating the performance of the AI model, and analyzing the effectiveness of the Federated Learning (FL) approach over the centralized approach. Data analysis was divided into three main components: AI model performance analysis, FL system efficiency analysis, and user feedback analysis.

3. RESULTS AND DISCUSSIONS

This section presents the results obtained from the development and evaluation of the health prediction application for house plants, which utilizes smartphone cameras and artificial intelligence (AI) integrated with Federated Learning (FL). The discussion highlights system performance, prediction accuracy, privacy preservation, and computational efficiency, followed by a comparative analysis with centralized learning approaches.

3.1. Model and Dataset Implementation

The machine learning model used is a Convolutional Neural Network (CNN), which is optimized for mobile devices using the MobileNetV2 architecture. The dataset used consists of 15,000 images of house plants, obtained through the voluntary participation of 200 smartphone users over 2 months. Each image was categorized into 6 classes, namely:

- a. Healthy
- b. Fungal infection
- c. Bacterial spot
- d. Pest infestation
- e. Nutrient deficiency
- f. Overwatering

The dataset is divided into 80% training data and 20% testing data, with variations in device, camera resolution, and lighting conditions that represent real field conditions.

The implementation of the health prediction model for house plants revolves around leveraging deep learning techniques, specifically convolutional neural networks (CNNs), to perform image classification and diagnosis of plant health status. CNNs were chosen due to their strong performance in image recognition tasks and their ability to extract hierarchical features from plant images, such as leaf color, texture, and shape-which are critical indicators of plant health. The architecture used in this research is based on a modified MobileNetV2 model, chosen for its balance between accuracy and computational efficiency. MobileNetV2 is lightweight and optimized for mobile and embedded devices, making it suitable for deployment in a smartphone-based application. The model consists of depthwise separable convolutions and inverted residual blocks, which significantly reduce the number of parameters while maintaining high accuracy.

The model outputs a multi-class classification, where each class corresponds to a specific plant health condition (e.g., healthy, nutrient deficiency, fungal infection, pest damage, etc.). A softmax

activation function is applied in the output layer to provide a probability distribution across classes. The training objective uses a categorical cross-entropy loss function, optimized via the Adam optimizer with a learning rate scheduler to enhance convergence. To support privacy-preserving learning, the model was implemented within a Federated Learning (FL) framework using TensorFlow Federated (TFF). This setup allows the model to be trained across multiple devices (clients) without transferring raw image data to a central server. Instead, each device computes model updates locally, which are then aggregated centrally using the Federated Averaging (FedAvg) algorithm. This decentralized learning process reduces privacy risks and bandwidth usage while enabling continuous model improvement. Security and efficiency in the FL setup were further reinforced through differential privacy mechanisms and model compression techniques. Differential privacy adds noise to the local updates to obscure specific user data, while model quantization ensures the FL communication overhead remains minimal.

The development of an effective and reliable model necessitates a diverse and well-annotated dataset. Given the scarcity of publicly available datasets tailored specifically for house plants, the research team constructed a custom dataset by collecting over 15,000 labeled images of common house plants under varying conditions. Images were gathered through crowd-sourcing, direct photography, and augmented with publicly available datasets such as PlantVillage and AI Challenger Crop Disease Dataset, after ensuring compliance with licensing terms and adjusting labels to suit the research scope. The implementation of the plant health prediction model using CNNs within a Federated Learning framework has demonstrated promising results in terms of performance, scalability, and user privacy. The use of MobileNetV2 allows real-time analysis on mobile devices, while FL ensures that users' image data remains secure. The custom dataset, tailored to the home environment and enriched with real-world diversity, plays a critical role in model robustness. Despite challenges related to data heterogeneity and system limitations, the application sets a strong foundation for scalable, privacy-preserving AI in smart home agriculture.

3.2. Model Evaluation and Accuracy

After the FL process was completed, the global model was evaluated with test data that had not been used before. The test results showed:

- a. Overall accuracy: 88,4%
- b. Average precision: 89,1%
- c. Average Recall: 87,7%
- d. Average F1-score: 88,4%

The best classification was found in the "Healthy" (93%) and "Fungal Infection" (91%) categories. However, there was a misclassification between "Nutrient Deficiency" and "Overwatering", possibly due to similar visual symptoms, such as yellowing and wilting of leaves.

Table 1. Evaluation Results Per Category

Metric	Federated Learning	Centralized Learning
Accuracy	88.4%	90.2%
Training Time	Longer	Faster
Privacy	Preserved	Not preserved
Data Transmission	Low	High
Device Computation	Required	Minimal

Although the centralized model yielded marginally higher performance, the FL-based approach offers a more sustainable and ethical alternative, especially when privacy and bandwidth are critical concerns.

For comparison, the same model was trained in a centralized manner using the entire dataset directly on a central server. Results:

- a. Accuracy of the centralized model: 90,2%
- b. Training speed: Faster
- c. Bandwidth consumption: Very high
- d. Data privacy: Not maintained

Meanwhile, the FL model shows efficiency in privacy and data consumption, despite the longer training time. This comparison shows that FL offers a good balance between accuracy, efficiency, and data security. To assess the performance of the health prediction model, a systematic evaluation was conducted using both centralized and federated learning settings. The model was evaluated across multiple metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis, with a

focus on its ability to correctly identify various health conditions in house plants from images taken using smartphone cameras. The evaluation dataset consisted of approximately 3,000 labeled images that were completely separated from the training set. This test set included multiple plant species, lighting conditions, and symptom types to replicate real-world diversity and ensure that the model generalizes well outside of its training distribution. In the federated learning scenario, the model was tested on multiple clients, each with local validation data representing unique, non-IID distributions. This setup aimed to reflect practical deployment conditions where each user may observe only a limited subset of plant types or conditions.

In the federated learning setup, the performance was evaluated after 100 rounds of communication, using 20 simulated clients, each with different subsets of the training data.

- a. Global Accuracy (aggregated model): 89.1%
- b. Precision: 88.3%
- c. Recall: 88.9%
- d. F1-Score: 88.6%

Although the federated model achieved slightly lower performance compared to the centralized model, it retained strong classification capabilities while maintaining user privacy. Importantly, the federated model was more robust to domain shifts, as it continued to improve over successive communication rounds. The accuracy gap (~3%) between centralized and federated approaches can be attributed to the non-IID nature of data among clients and partial participation (i.e., not all clients contribute in every round). Nonetheless, the drop is considered acceptable given the privacy and scalability advantages gained.

3.3. Field Trial and User Feedback

After the training was completed, the app was tested by 50 users for 14 days. They were asked to take pictures of their plants and receive diagnosis results from the app.

- a. User survey results
- b. Satisfaction level: 86% satisfied with the diagnosis results
- c. Average response time: 2.3 seconds per prediction
- d. Perceived accuracy: 82% felt the diagnosis matched reality
- e. Trust in privacy: 91% felt safer using the system without image uploads

This trial shows that the system works well in real-life conditions, even on mid-range phones.

To evaluate the real-world performance and usability of the developed mobile application, a field trial was conducted involving 50 voluntary participants over a 14-day period. Participants were selected to represent a broad demographic, including plant enthusiasts, casual users, and urban farmers. Each participant installed the application on their personal Android smartphone, used the app to capture images of their house plants, and received instant feedback regarding the plant's health condition.

The primary objectives of the trial were to assess:

- a. Diagnostic accuracy from the user's perspective
- b. Application responsiveness and performance under diverse device and environmental conditions
- c. User experience (UX) and satisfaction
- d. Perceived trust and confidence in data privacy and security

All user interactions with the app were logged (with user consent), and a post-trial survey was administered to collect qualitative and quantitative feedback.

During the trial, the application was used to analyze a total of 2,436 plant images across different lighting conditions, camera types, and plant species. The following observations were recorded: Average response time: 2.3 seconds per prediction (measured from image capture to displaying results) Failure rate: 3.7%, mostly due to poor image quality (e.g., blur or insufficient lighting), Battery consumption: Moderate (an estimated 4–6% per 10 image evaluations) Model synchronization (federated updates): Successfully completed by 87% of users during the trial period, These metrics confirm the system's ability to operate efficiently on a wide range of smartphones, maintaining reasonable computational and energy demands.

The field trial results confirm the technical feasibility and user acceptance of the application in real-life settings. The high rate of perceived diagnostic accuracy, fast response time, and strong privacy sentiment demonstrate the potential of Federated Learning to power intelligent, user-respecting solutions in consumer-facing domains. While centralized AI systems often face skepticism due to data privacy issues, this study shows that decentralized approaches like FL can provide comparable

performance while enhancing user trust. Moreover, the mobile-first design aligned well with users' daily routines, enabling natural integration into plant care habits. However, limitations such as occasional misclassifications and system dependency on image quality highlight the need for: Improved preprocessing on-device, A broader dataset encompassing more plant types and diseases, Contextual data integration (e.g., humidity, season, location) for more holistic prediction, From a research standpoint, the trial reinforces that usability testing and participatory design are critical components in the successful deployment of AI applications beyond controlled laboratory settings.

4. CONCLUSION

This study successfully developed and evaluated a mobile-based application that leverages artificial intelligence (AI) and Federated Learning (FL) to predict the health condition of house plants using images captured through smartphone cameras. The integration of FL into the application architecture addresses major concerns around data privacy, allowing model training to occur locally on user devices without the need to transmit sensitive image data to centralized servers. The application employs a lightweight convolutional neural network (CNN) model optimized for mobile environments. It demonstrated high accuracy in classifying plant health conditions across multiple categories, including healthy, fungal infection, pest infestation, overwatering, and nutrient deficiencies. Through iterative training across devices and federated aggregation, the global model achieved an accuracy of 88.4%, which is comparable to centralized approaches while preserving user privacy. Field trials involving 50 participants over two weeks revealed strong user satisfaction, with 82% agreeing with the app's predictions and 91% expressing trust in the privacy-preserving features. The system also performed reliably across a variety of smartphones, achieving an average response time of 2.3 seconds per prediction and maintaining a low failure rate. Despite its promising performance, the application faced challenges such as sensitivity to image quality and difficulty distinguishing between symptoms with similar visual features. User feedback suggested enhancements such as more detailed care recommendations, plant species input, and improved offline support. In conclusion, this study demonstrates the potential of combining mobile AI and Federated Learning to create smart, privacy-aware plant care tools. The results affirm that such systems are not only technically feasible but also well-received by end users. Future work will focus on improving model robustness, expanding plant species coverage, and incorporating multi-modal data to enhance diagnostic accuracy and user experience.

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